

Name:

Rollno:

CS345: Design and Analysis of Algorithms (Quiz 1)

Total Number of Pages: 4

Time: 1hr

Total Points 40

Instructions

1. All questions are compulsory.
2. Answer all the questions in the space provided in the question paper booklet.
3. Use the space provided in the paper for rough work.
4. The symbols or notations mean as usual unless stated.
5. You may cite and use algorithms and their complexity as done in the class.
6. Cheating or resorting to any unfair means will be severely penalized.
7. Superfluous and irrelevant writing will result in negative marking.
8. Use blue/black ink and not pencils. Do not use red pens.

Question	Points	Score
1	3	
2	3	
3	3	
4	3	
5	3	
6	3	
7	3	
8	3	
9	3	
10	3	
11	10	
Total:	40	

Helpful hints

1. It is advisable to solve a problem first before writing down the solution.
 2. The questions are *not* arranged according to increasing order of difficulty.
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Question 1. (3 points) Consider the recurrence relation

$$T(n) = \sqrt{n}T(\sqrt{n}) + n.$$

Determine the asymptotic time complexity of $T(n)$.

Answer: _____ $O(n \log \log n)$ _____

Question 2. (3 points) When using Divide and Conquer to find non-dominated points (maxima) in a 2D set sorted by x -coordinate, the set is split by a vertical line into a left half L and a right half R . How do the non-dominated points of L and R combine?

- A. Points in R discard points in L that have a lower y -coordinate than the highest point in R .
- B. Points in L discard points in R that have a lower x -coordinate.
- C. The merge step requires $O(n^2)$ comparisons.
- D. All non-dominated points in L and R remain non-dominated.

Question 3. (3 points) We are given 9 tasks $T_1, T_2, \dots, T_9$. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i . Profit P_i is earned if the task is completed before the end of the d_i -th unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

What is the maximum profit earned?

- A. 147
- B. 165
- C. 167
- D. 175

Question 4. (3 points) The characters a to h have frequencies as follows:

$$a : 1, \quad b : 1, \quad c : 2, \quad d : 3, \quad e : 5, \quad f : 8, \quad g : 13, \quad h : 21.$$

A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the code

110111100111010?

- A. fdheg
- B. ecgdf
- C. dchfg
- D. fehdg

Question 5. (3 points) Suppose a DFS is performed on a directed graph and let (u, v) be an edge in the graph. Then which of the following relations between the discovery and finish time of the vertices is NOT possible?

- A. $D[u] < D[v] < F[v] < F[u]$
- B. $D[v] < D[u] < F[u] < F[v]$
- C. $D[u] < F[u] < D[v] < F[v]$

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D. $D[v] < F[v] < D[u] < F[u]$

Question 6. (3 points) Let G be a directed graph. Consider a depth-first traversal of G , and let T be the resulting DFS forest. Let u be a vertex in G , and let v be the first new (unvisited) vertex discovered after u during the traversal.

Which of the following statements is always true?

- A. (u, v) must be an edge in G , and v is a descendant of u in T .
- B. (v, u) must be an edge in G , and u is a descendant of v in T .
- C. (u, v) must be a forward edge.
- D. If (u, v) is not an edge in G , then u and v must have the same parent in T .

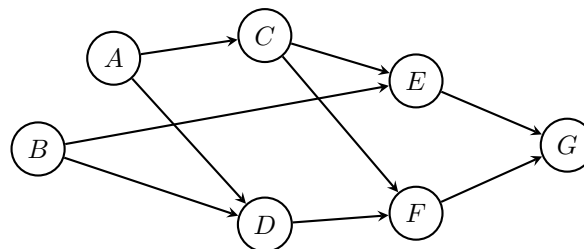
Question 7. (3 points) Let T be a depth-first search tree of a directed graph G . Vertices u and v are leaves of T . Suppose the out-degree of both u and v in G is at least 1. Which one of the following statements is always true?

- A. There must exist a vertex w such that both u and v have an edge directed towards w .
- B. Every outgoing edge from u and every outgoing edge from v is either a back edge or a cross edge in T .**
- C. There must exist a directed cycle in G containing both u and v .
- D. There must exist a directed cycle in G containing u and all vertices reachable from u by one outgoing edge.

Question 8. (3 points) Let G be a unique path graph. Which of the following statements are true?

- A. DFS tree from some vertex can have forward edges.
- B. DFS tree from some vertex can have more than one backward edges.**
- C. DFS tree from some vertex can have more than one backward edge going out of a vertex in the tree
- D. DFS forest of the graph may have cross edges.**

Question 9. (3 points) Consider the following directed acyclic graph.



Which of the following is **NOT** a valid topological ordering?

- A. A, B, C, D, E, F, G
- B. B, A, C, D, E, F, G
- C. A, C, B, E, D, F, G
- D. A, C, E, B, D, F, G**

Question 10. (3 points) During an interval tree search, if the maximum high endpoint of all intervals in the left subtree L is strictly less than $\text{Low}(I)$, the algorithm safely prunes the left subtree and moves to the right child. This is because every interval stored within L is mathematically guaranteed to lie entirely to the left of the query interval I on the number line.

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Question 11. (10 points) Given an array that represents elements of a geometric progression in order. One element is missing in the progression. Design an algorithm to find the missing number in $O(\log n)$ time.

Solution: Let the array be $A[0 \dots n-1]$, containing the terms of a geometric progression in order, with exactly one term missing.

Key Idea. Suppose the common ratio of the geometric progression is r . Then, ideally,

$$A[i] = A[0]r^i.$$

Since one element is missing, the original geometric progression has $n+1$ terms. Therefore,

$$A[n-1] = A[0]r^n.$$

Thus,

$$r = \left(\frac{A[n-1]}{A[0]} \right)^{1/n}.$$

We can now use binary search. For any index mid , the expected value is

$$\text{expected} = A[0]r^{mid}.$$

- If $A[mid]$ equals the expected value, then the missing element lies to the right.
- Otherwise, the missing element lies to the left, and the expected value at the first mismatch is the missing element.